Freesurfer data - Insight46 phase 1 and 2

Sarah Keuss | September 2022

Contents

Document details

Output variables	lh_adm_corthck_i46p1, rh_adm_corthck_i46p1, lh_adh_corthck_i46p1, rh_adh_corthck_i46p1, mean_adm_corthck_i46p1, mean_adh_corthck_i46p1, lh_adm_corthck_i46p2, rh_adm_corthck_i46p2, lh_adh_corthck_i46p2, rh_adh_corthck_i46p2, mean_adm_corthck_i46p2, mean_adh_corthck_i46p2

Overview

Insight 46 phase one and two Freesurfer data

Methods

T1 MRI data from each time-point underwent correction for gradient non-linearity (Jovicich et al., 2006), and were manually checked by a trained team who assessed motion, anatomical coverage, and other quality control issues. Scan-pairs were also reviewed to check for unacceptable differences in quality between time-points. A quality control summary is saved on Icarus in the following location: 1946\Test_Data_and_Video_Files\Phase_2\Imaging Data\Phase2_Data_Freeze\Insight46_PHASE2-FREEZE-20201124_QC_SUMMARY.xlsx.

Phase 1 T1 MRI data that passed cross-sectional quality control (n=468) and phase 2 T1 MRI data that passed cross-sectional and longitudinal quality control (n=356) were processed using Freesurfer 7.1.0 (). Images from each time-point first underwent cross-sectional processing, including motion correction, removal of the skull, spatial transforms, intensity normalisation, cortical surface reconstruction and parcellation (Dale, Fischl and Sereno, 1999; Fischl, Sereno and Dale, 1999; Fischl and Dale, 2000; Desikan et al., 2006). Cross-sectionally processed images were then run through the longitudinal stream in Freesurfer (Reuter et al., 2012). Specifically, an unbiased within-subject template was created based on data from both time-points using robust inverse, consistent registration (Reuter, Rosas and Fischl, 2010; Reuter and Fischl, 2011), and then several processing steps were initialised using common information from this template (Reuter et al., 2012).

All cortical thickness segmentations were visually inspected for major errors.

This work was completed by Sarah Keuss () and Will Coath () with senior guidance from Dave Cash (). An Insight 46 paper which will summarise these methods is currently in progress.

References

Dale, A. M., Fischl, B. and Sereno, M. I. (1999) 'Cortical Surface-Based Analysis: I. Segmentation and Surface Reconstruction', NeuroImage, 9(2), pp. 179–194. doi: 10.1006/NIMG.1998.0395.

Desikan, R. S. et al. (2006) 'An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest', NeuroImage, 31(3), pp. 968–980. doi: 10.1016/j.neuroimage.2006.01.021.

Fischl, B. and Dale, A. M. (2000) 'Measuring the thickness of the human cerebral cortex from magnetic resonance images', Proceedings of the National Academy of Sciences, 97(20), pp. 11050–11055. doi: 10.1073/pnas.200033797.

Fischl, B., Sereno, M. I. and Dale, A. M. (1999) 'Cortical Surface-Based Analysis: II: Inflation, Flattening, and a Surface-Based Coordinate System', NeuroImage, 9(2), pp. 195–207. doi: 10.1006/NIMG.1998.0396.

Jovicich, J. et al. (2006) 'Reliability in multi-site structural MRI studies: Effects of gradient non-linearity correction on phantom and human data', NeuroImage, 30(2), pp. 436–443. doi: 10.1016/j.neuroimage.2005.09.046.

Reuter, M. et al. (2012) 'Within-subject template estimation for unbiased longitudinal image analysis.', NeuroImage, 61(4), pp. 1402–18. doi: 10.1016/j.neuroimage.2012.02.084.

Reuter, M. and Fischl, B. (2011) 'Avoiding asymmetry-induced bias in longitudinal image processing', NeuroImage, pp. 19–21. doi: 10.1016/j.neuroimage.2011.02.076.

Reuter, M., Rosas, H. D. and Fischl, B. (2010) 'Highly accurate inverse consistent registration: A robust approach', NeuroImage, 53(4), pp. 1181–1196. doi: 10.1016/j.neuroimage.2010.07.020.

Freesurfer data storage

All output files are currently zipped and stored on the UCL. Please liaise with Will Coath () and Dave Cash () if access to this data is required.

Data requests

Sarah Keuss has shared AD signature cortical thickness variables with the LHA, and they will be accessible via the usual route on Skylark in due course. They are also stored on Icarus in the following location: 1946\Test_Data_and_Video_Files\Phase_2\Imaging Data\Phase2_Data_Freeze\Insight46_phase1&2_ADsigcorticalthickness_cleaned_final_20221024.xlsx

For other variables/analyses involving Freesurfer, the current working plan is:

- Internal UCL researchers – to be given direct access to the output files via the cluster or to copy the files they require into their own folder on the cluster.
- External researchers – TBC (discuss with Dave Cash and Jon Schott); one option is to use the XNAT server and attach the zipped output files to the shared imaging data.

Anyone requesting data should be given the methods summary above. A separate datasheet describing how AD signature cortical thickness variables were derived is saved on Icarus in the following location: 1946\Test_Data_and_Video_Files\Phase_2\Imaging Data\Phase2_Data_Freeze\Insight46_phase1&2_ADsigcorticalthickness_cleaned_final_20221024.docx.

Variables in this document

12 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › Cortical thickness [60334]		
lh_adh_corthck_i46p1	Left hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
lh_adh_corthck_i46p2	Left hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
lh_adm_corthck_i46p1	Left hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
lh_adm_corthck_i46p2	Left hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
mean_adh_corthck_i46p1	Surface area weighted average of left and right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
mean_adh_corthck_i46p2	Surface area weighted average of left and right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
mean_adm_corthck_i46p1	Surface area weighted average of left and right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
mean_adm_corthck_i46p2	Surface area weighted average of left and right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field

Variable	Label	How it was found
rh_adh_corthck_i46p1	Right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
rh_adh_corthck_i46p2	Right hemisphere ADsig Harvard cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field
rh_adm_corthck_i46p1	Right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 1	topsheet field
rh_adm_corthck_i46p2	Right hemisphere ADsig Mayo cortical thickness (mm) - Longitudinal - Insight46 phase 2	topsheet field

1 low-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Sociodemographics [515] › Household [5152]		
own	Now I would like to get some general information about your household. Does your household own or rent this accommodation? - at age 53 years	prose scan